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PAPER_1088: $F_{U,Bi,i}$ Seven-Component Force Decomposition

Abstract

The unified buoyancy force $F_{U,Bi,i}$ is decomposed into seven orthogonal sectors that account for all fundamental buoyancy-mediated interactions in the UQFF framework:

$$F_{U,Bi,i} = F_{\text{phonon}} + F_{\text{inflation}} + F_{\text{BCS}} + F_{\text{VDS}} + F_{\text{DVP}} + F_{\text{BSH}} + F_{\text{QCalcGeom}}$$

Each sector contributes a fractional share to the total buoyancy budget measured against the base gravitational acceleration $g_{\text{extbase}} = \mu_s \nabla(M_s/r)$.

§1 Base Gravity and Component Definitions

$$g_{\text{base}} = \frac{GM}{r^2}, \quad G = 6.674 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$$

The seven force components are:

§1.1 Phonon Sector

$$F_{\text{phonon}} = \Phi_{1.25 \text{ THz}} \cdot g_{\text{base}} \cdot M$$

Vacuum phonon resonance at 1.25 THz, coupling buoyancy to the SCm lattice oscillation spectrum.

§1.2 Inflation Sector

$$F_{\text{inflation}} = \beta_i \cdot U_g \cdot \frac{M}{d} \cdot [\text{UA}]$$

Buoyancy force driving inflationary expansion, with $\beta_i = 0.603$ and $[\text{UA}] = 10^{-4}$.

§1.3 BCS (Buoyancy-Condensate-Superfluid) Sector

$$F_{\text{BCS}} = \Delta_{\text{BCS}}^2 \cdot g_{\text{base}} \cdot M$$

Cooper-pair condensate contribution from the BCS gap parameter Δ_{BCS} .

§1.4 VDS (Vacuum Density Series) Sector

$$F_{\text{VDS}} = \rho_{\text{vac}} \cdot V_{\text{eff}} \cdot g_{\text{base}}$$

Vacuum density series contribution encoding $k_\eta = 10^{-113}$.

§1.5 DVP (Dipole Vortex Primes) Sector

$$F_{\text{DVP}} = \sum_{p \in \mathcal{P}} \frac{\mu_p}{r_p^3} \cdot g_{\text{base}} \cdot M$$

Vortex-line contributions indexed by prime-number dipole configurations.

§1.6 BSH (Buoyancy Shell Harmonics) Sector

$$F_{\text{BSH}} = \sum_{\ell=0}^{26} Y_{\ell}^0(\theta) \cdot g_{\text{base}} \cdot M$$

Spherical-harmonic shell decomposition across 26 layers.

§1.7 QCalcGeom (Quantum Calculator Geometry) Sector

$$F_{\text{QCalcGeom}} = \alpha_{\text{QCG}} \cdot g_{\text{base}} \cdot M$$

Geometric correction from the quantum calculator manifold structure.

§2 Fractional Contribution Table

Sector	Force Component	Fraction of $F_{U,Bi,i}$
Phonon	F_{phonon}	Dominant at resonance
Inflation	$F_{\text{inflation}}$	$\beta_i / \sum$
BCS	F_{BCS}	$\sim 10^{-3}$
VDS	F_{VDS}	$\sim 10^{-5}$
DVP	F_{DVP}	$\sim 10^{-4}$
BSH	F_{BSH}	$\sim 10^{-2}$
QCalcGeom	$F_{\text{QCalcGeom}}$	$\sim 10^{-3}$

§3 Budget Closure

$$\sum_{k=1}^7 f_k = 1, \quad f_k \equiv \frac{F_k}{F_{U,Bi,i}}$$

The seven fractions are computed dynamically per system and must satisfy exact budget closure to machine precision.

References

- PAPER_1089: Inflation Buoyancy Sector Lagrangian
- PAPER_1090: Dark Energy Buoyancy Sector Lagrangian
- PAPER_892: FUBiUnifiedFieldBuoyancyForceCalc
- PAPER_893: GravBuoyancyForceCalculator

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Archimedes (~250 BCE). *On Floating Bodies*. (Principle of buoyancy)

4. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — arXiv:astro-ph/0004212
5. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x
6. Anderson, M.H. et al. (1995). *Observation of Bose-Einstein Condensation in a Dilute Atomic Vapor*. Science **269**, 198 — doi:10.1126/science.269.5221.198
7. Dalfovo, F. et al. (1999). *Theory of Bose-Einstein condensation in trapped gases*. Rev. Mod. Phys. **71**, 463 — arXiv:cond-mat/9806038 — doi:10.1103/RevModPhys.71.463
8. Pitaevskii, L. & Stringari, S. (2003). *Bose-Einstein Condensation*. Oxford: Clarendon Press
9. Ashcroft, N.W. & Mermin, N.D. (1976). *Solid State Physics*. Harcourt
10. Kittel, C. (2004). *Introduction to Solid State Physics*. 8th ed. Wiley
11. Feynman, R.P. (1982). *Simulating Physics with Computers*. Int. J. Theor. Phys. **21**, 467 — doi:10.1007/BF02650179
12. Guth, A.H. (1981). *Inflationary universe: A possible solution to the horizon and flatness problems*. Phys. Rev. D **23**, 347 — doi:10.1103/PhysRevD.23.347
13. Starobinsky, A.A. (1980). *A new type of isotropic cosmological models without singularity*. Phys. Lett. B **91**, 99 — doi:10.1016/0370-2693(80)90670-X
14. BICEP/Keck Collaboration (2021). *Improved Constraints on Primordial Gravitational Waves using Planck, WMAP, and BICEP/Keck Observations*. Phys. Rev. Lett. **127**, 151301 — arXiv:2110.00483 — doi:10.1103/PhysRevLett.127.151301